



**Ghulam Ishaq Khan Institute of Engineering Sciences and
Technology**

AI201

Project Proposal

**Interactive Spectrographic Modeling
and AI Classification of Bird Calls**

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Date: November 29, 2025

1. Project Title

Interactive Spectrographic Modeling and AI Classification of Bird Calls

2. Problem Statement

Bird vocalizations provide essential information for biodiversity and ecosystem monitoring. Manual identification of species from audio recordings is slow, inconsistent, and requires specialized expertise.

Existing automated tools primarily focus on *classification* while offering limited support for deeper *visual analysis*. This project aims to develop a system that combines species classification with interactive 3D visualizations of bird vocal signatures, supporting researchers, conservationists, and educators.

3. Objectives

1. Collect and organize bird audio datasets.
2. Extract audio features such as MFCCs, spectrograms, and chromagrams.
3. Design interactive 3D visualizations of audio features.
4. Develop an optional species classification model.
5. Build a simple interface for audio upload and visualization.

4. Methodology

1. **Data Collection:** Recordings from Xeno-canto, Cornell Lab, Kaggle.
2. **Preprocessing:** Normalization, noise reduction, segmentation.
3. **Feature Extraction:** Compute spectrograms, MFCCs, chromagrams using Librosa.
4. **3D Representation:** Map features onto 3D space (frequency–time–amplitude).
5. **Visualization Tools:** Plotly, Matplotlib 3D, or Mayavi.
6. **Machine Learning (Optional):** Use Scikit-learn or PyTorch for species classification.
7. **Interface:** Provide a simple dashboard for uploading audio files.

5. Expected Outcomes

- A well-organized and preprocessed bird audio dataset.
- Interactive 3D visualizations for exploring sound patterns.
- A functional species classifier (optional).
- Complete documentation of the data processing and modeling pipeline.

6. Tools and Libraries

Programming Language: Python (with OOP structure)

Audio Processing: Librosa, NumPy

Data Handling: Pandas

Visualization: Plotly, Matplotlib, Mayavi

Machine Learning: Scikit-learn, PyTorch/TensorFlow (optional)

Version Control: Git/GitHub

7. Checklist Mapping

Requirement	Status
Exception Handling & Version Control	Included
Object-Oriented Programming (OOP)	Used for modules
NumPy	Used in feature extraction
Pandas	Used for metadata management
3D Visualization (Matplotlib)	Implemented
PyTorch / TensorFlow	Optional
Scikit-learn	For ML models
External Libraries	Librosa, Plotly, etc.

8. Potential Usefulness

- Supports biodiversity and conservation research.
- Enables detection of changes in species behavior.
- Useful for education, citizen science, and ecological monitoring.
- Demonstrates practical integration of audio processing, AI, and visualization.